

Triasha Sarkar

Machine Learning Engineer | Ex-Rolls-Royce · United States · tsarkar34@gatech.edu · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

SUMMARY

Machine learning engineer with aerospace systems experience building evaluation pipelines, retrieval systems, surrogate models, and deployment evidence. I work across data contracts, model evaluation, reproducible experiments, and production boundaries. My focus is making ML results measurable, reviewable, and useful to engineering teams.

EXPERIENCE

Machine Learning Engineer — Rolls-Royce Jul 2023 – Apr 2025

- Built machine-learning and predictive-maintenance workflows for aircraft-engine sensor data.
- Developed Python pipelines for 1 Hz and 10 Hz multivariate time-series data, including cleaning, feature engineering, anomaly detection, and model evaluation.
- Worked with Lifecycle Engineers and domain experts to understand failure modes and turn model outputs into maintenance insights.
- Used Azure Databricks for repeatable analysis of large engineering datasets and contributed to solutions supporting earlier issue detection.

Graduate Research Assistant — Aerospace Systems Design Laboratory, Georgia Tech May 2025 – Aug 2026

- Built a failure-detection workflow using 15,120 simulated rocket-motor time-series cases, reaching 96.8% accuracy and 0.986 ROC AUC on unseen geometries.
- Worked on the Delta Air Lines-sponsored HERO project, exploring retrieval and RAG methods for safety-related information.
- Developed surrogate models for the GREEN TEA project, connecting lifecycle, economic, and transportation data for faster engineering analysis.
- Applied Python, scikit-learn, statistical modeling, feature engineering, clustering, and model evaluation with a focus on reproducibility and failure analysis.

Project EAGLE — Vehicle Grand Challenge with RIC Enterprise, Georgia Tech 2025 – 2026

Worked on Project EAGLE (Eco-friendly Airborne Global Luxury Ekranoplan), a two-semester industry-sponsored research project. Developed a sizing and synthesis tool for a one-of-a-kind vehicle, and developed the passenger-accommodation layout and cabin geometry in AutoCAD while modifying and developing vehicle requirements.

Project GREEN TEA — System of Systems Grand Challenge, Georgia Tech 2025 – 2026

Worked on Project GREEN TEA (Generalized Residue to Energy Evaluation Network and Techno-Economic Analysis), an industry-sponsored project with the U.S. Endowment for Forestry and Communities. Developed the architecture and framework for a techno-economic tool analyzing sustainable aviation fuel and bioenergy from forest residues, including jobs, CO₂ impacts, a GREET life-cycle-analysis surrogate model, and a forest-health impact model.

Intern — STAR Orbitals Aug 2021 – Oct 2021

Interned at STAR Orbitals from Aug 2021 to Oct 2021, contributing to static-test-pad development for solid rocket motors, client timelines, project planning, Arduino avionics, SolidWorks, and Autodesk Fusion 360 mechanical design.

Intern — Rolls-Royce DataLabs May 2021 – Jul 2021

Worked on aerospace data projects involving predictive analytics, anomaly detection, model-based engineering, exploratory analysis, feature engineering, and time-series sensor data in a safety-critical environment.

Summer Intern — Bangalore Aircraft Industries (BAIL) Jun 2019 – Jul 2019

Worked on material testing, stress analysis, and structural testing using NASTRAN-PATRAN.

SELECTED ENGINEERING PROJECTS

AeroRAG-X — Retrieval and agent evaluation [GitHub](#)

Built evaluation contracts for retrieval, citation support, agent trajectories, safe degradation, observability, and held-out comparisons. Includes QASPER/SciFact evaluation paths and citation-corruption tests.

NewsLens — Stateful news retrieval service [GitHub](#)

Built a containerized retrieval service with separate application and state hosts, Kafka, Postgres, health checks, evidence runners, and documented deployment boundaries.

IntegrityBench — Integrity and safety evaluation[GitHub](#)

Built fail-closed release gates, public evidence receipts, CUDA transformer evaluation paths, and candidate assessment tooling without overstating missing human-review evidence.

EdgeGenBench — Edge inference evidence[GitHub](#)

Built reference and QNN evidence validators, generated-data aircraft design experiments, runtime contracts, and device-evidence capture workflows.

SELECTED ENGINEERING PROJECTS (CONTINUED)

AeroSynth-Eval — Generative-model evaluation[GitHub](#)

Built preference, calibration, transfer, downstream-utility, grouped-uncertainty, and human-alignment evaluation contracts. Synthetic and real-data claims are kept separate.

Equity-Backtest — Leakage-aware research pipeline[GitHub](#)

Built a point-in-time backtesting pipeline with membership validation, leakage checks, turnover, capacity constraints, and explicit boundaries around licensed CRSP data.

Surrogate Model Learning — Robust regression[GitHub](#)

Built Gaussian-process and regression workflows for engineering data, repeated grouped robustness reports, uncertainty checks, and operating-domain warnings.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, MATLAB, Bash

ML: PyTorch, scikit-learn, Gaussian processes, regression, retrieval, ranking, evaluation, uncertainty, time series

Systems: Docker, Kafka, Postgres, AWS, CI, ONNX, QNN, REST APIs, observability, reproducibility contracts

Engineering: requirements management, systems architecture, verification and validation, risk analysis, model cards, technical documentation

RESEARCH AND ENGINEERING FOCUS

- Scientific machine learning and surrogate modeling for aerospace and energy systems.
- Evaluation of retrieval, generation, safety, citation support, and model integrity.
- Reliable ML systems with explicit data, deployment, and evidence boundaries.

EARLIER AEROSPACE PROJECTS

- **GKN Aerospace hybrid aircraft design:** thermal-management integration for a zero-emission demonstrator using hybrid, electric, and hydrogen system concepts.
- **Supersonic business-jet thesis:** studied primary-nozzle geometry and noise-mitigation constraints for take-off and landing.
- **Bird-strike structural analysis:** used ANSYS to compare nose-cone material response under impact loading.

EDUCATION

M.S., Aerospace Engineering — Georgia Institute of Technology2026

M.Sc., Aerospace Vehicle Design — Cranfield University2023

B.Tech., Aerospace Engineering — SRM Institute of Science and Technology2021